



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS ADVANCED

2020 Year 12 Course Assessment Task 1

Monday, 2 December 2019

General instructions

- Working time – 1 hour (plus 5 minutes reading time).
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt all questions.
- All necessary working should be shown in every question.
- Marks may be deducted for illegible or incomplete working.
- A NESA Reference Sheet is provided.

NESA STUDENT #:

Class: (please ✓)

☐ 12MAT.1 – Ms Ham

☐ 12MAT.2 – Mr Lam

☐ 12MAT.3 – Mr Sekaran

☐ 12MAT.4 – Ms Nam

☐ 12MAT.5 – Mrs Gan

☐ 12MAT.6 – Ms Park

Marker's use only

QUESTION	1	2	3	Total	%
MARKS	$\overline{11}$	$\overline{18}$	$\overline{24}$	$\overline{53}$	

Question 1 (11 marks)**Marks**(a) Differentiate with respect to x :

i. $y = (3 + e^x)^5$

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ii. $y = \frac{e^{2x}}{2x+1}$

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(b) Find the equation of the tangent to $y = e^{-x}$ at the point (0,1).**2**

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- (c) Sketch the graph $y = 2e^{-2x} - 2$, showing all important features.

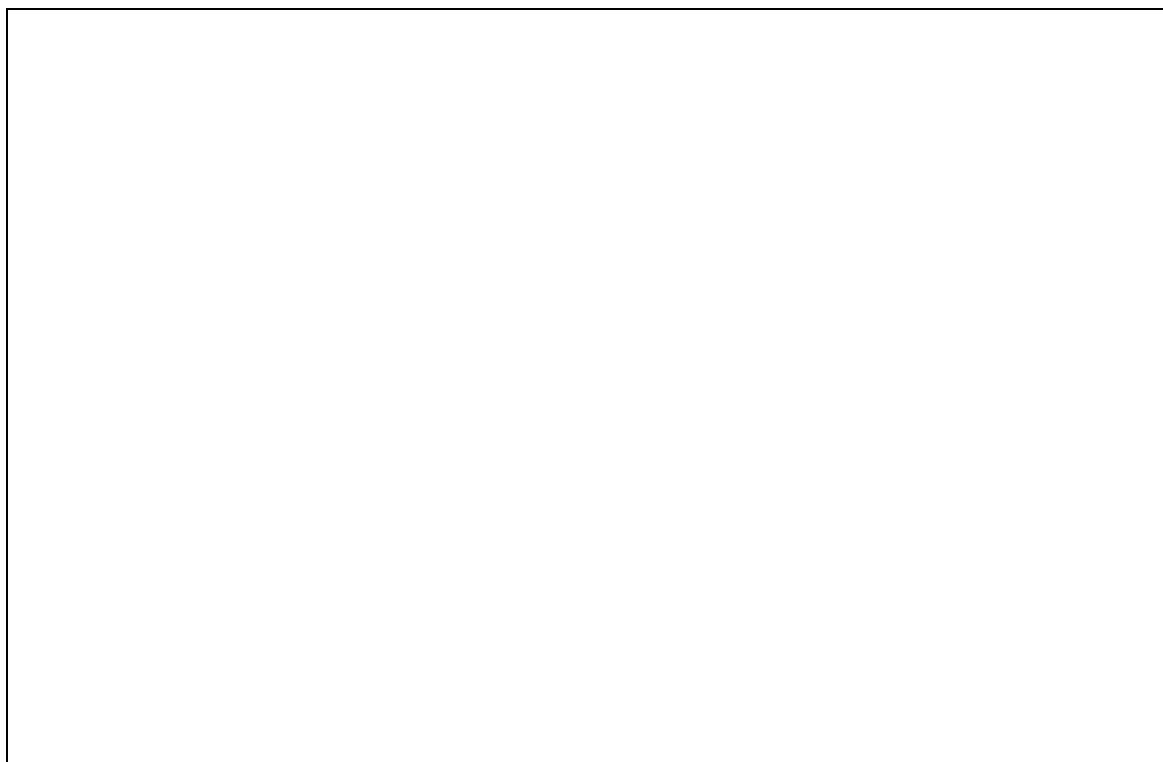
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- (d) Let $f(x) = e^x + k$, where k is a real number. The tangent to the graph of $y = f(x)$ at the point where $x = a$ passes through the point $(0, 0)$.

i. Show that the equation of the tangent is $y = e^a x$.

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ii. Find the value of k in terms of a .

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Question 2 (18 marks)**Marks**

- (a) Evaluate the following:

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$$\sum_{k=2}^{10} 4 \times \left(\frac{1}{2}\right)^{k-1}$$

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- (b) A geometric series has common ratio
- $\frac{1}{p}$
- and limiting sum
- $\frac{1}{1-p}$
- .

2Find the first term of this series, a , in simplest form.

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- (c) $2m - 8$, $2m + 4$ and $5m - 2$ are successive terms of a geometric sequence. Find the value of m . **3**

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- (d) i. Show that $\log_{10} 27 + \log_{10} 9 + \log_{10} 3 + \dots$ is an arithmetic series. **1**

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- ii. Show that the sum of the first seven terms of this series is 0. **2**

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- (e) A pattern is made up of a series of concentric circles with centre O . The radius of the first (innermost) circle is 10 cm. The distance between two adjacent concentric circles is 2 cm.

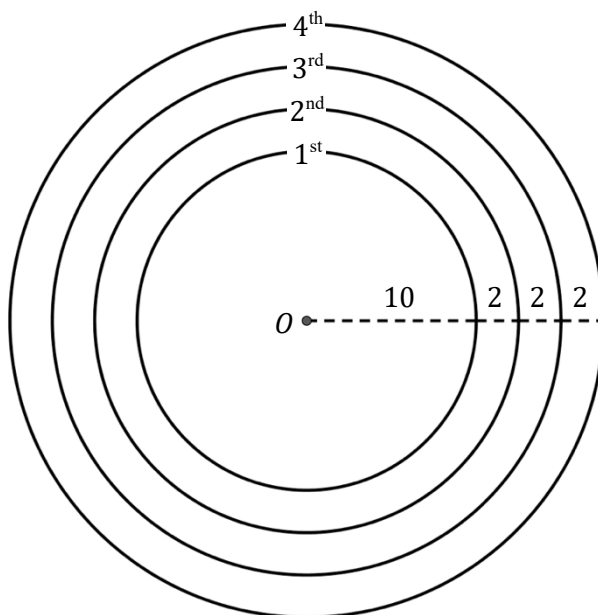


Diagram not drawn to scale

- i. What is circumference of the first circle?

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- ii. Show that the circumference, C , of the n^{th} circle is given by $C = 4\pi(4 + n)$.

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- iii. Show that the total circumference of n circles is given by $L = 2\pi n(9 + n)$.

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- iv. If $L = 1584\pi$, find the value of n and hence find the radius of the n^{th} circle.

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Question 3 (24 marks)**Marks**

- (a) 8 members of a rowing team were on a training row.

During the training row 6 members of the team were sunburnt and 2 were sunburnt and bitten by sand-flies. 1 member had nothing happen to him.

- i. Draw a Venn diagram showing this information.

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- ii. Hence find the probability that he was sunburnt but not bitten by sand-flies if a student was selected at random from the rowing team.

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- (b) The chance Janine can solve a probability problem is $\frac{9}{10}$ and the chance Naomi can solve it is $\frac{4}{5}$. If they work on the problem independently, what is the probability that:

- i. They both solve the problem?

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- ii. At least 1 of them is able to solve the problem?

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- (c) Four identical balls are numbered 1, 2, 3 and 4 and put into a box.

A ball is randomly drawn from the box, and not returned to the box. A second ball is then randomly drawn from the box.

By drawing a diagram or otherwise, find the probability that:

- i. The first ball drawn is numbered 4 and the second ball drawn is numbered 1. 1

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- ii. The sum of the numbers on the two balls is 5. 1

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- iii. The second ball drawn is numbered 1, given that the sum of the numbers on the two balls is 5. 2

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(d) A and B are events of a sample space. Given that $P(A|B) = p$, $P(B) = p^2$ and $P(A) = p^{\frac{1}{3}}$:

i. Show that $P(A \cap B) = p^3$.

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ii. Find $P(B|A)$ in simplest form.

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(e) From a packet of mixed seed, it was estimated that the probability of any seed planted yielding a white carnation was 0.02.

- i. Calculate the probability that from any two seeds planted there will be at least 1 white carnation. 2

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- ii. How many seeds must be planted to be at least 95% certain of obtaining at least 1 white carnation? 3

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- (f) Reung has invented a game for one person. He throws 2 ordinary dice repeatedly until the sum of the two numbers shown is either a 7 or 9.

If the sum is 9, Reung wins. If the sum is 7, Reung loses. If the sum is any other number, he continues to throw until it is a 7 or 9.

- i. Show that the probability that Reung wins on his first throw is $\frac{1}{9}$.

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- ii. Calculate the probability that a second throw is needed for Reung to win.

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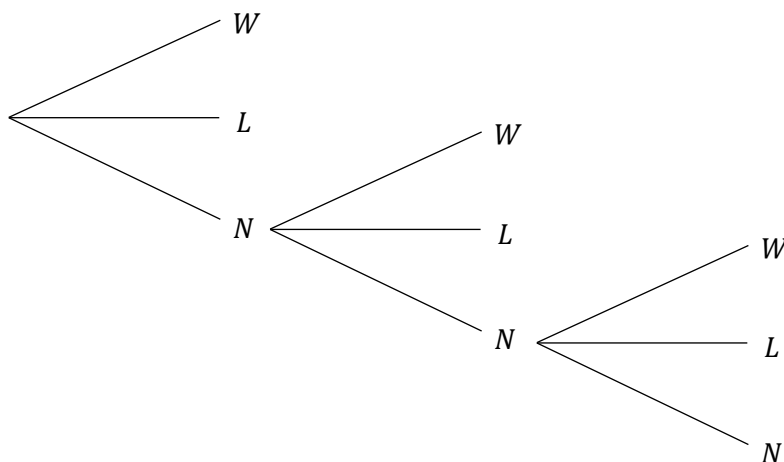
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Examination continues overleaf...

- iii. By completing the tree diagram or otherwise, find the probability that Reung wins on his first, second or third throw.

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- iv. Calculate the probability that Reung wins the game.

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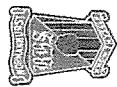
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End of Examination

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SOLUTIONS



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2020 Year 12 Course Assessment Task 1

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QUESTION	1	2	3	Total	%
MARKS	11	18	24	53	

Question 1 (11 marks)

Marks

(a) Differentiate with respect to x :

i. $y = (3 + e^x)^5$

2

$$\frac{dy}{dx} = 5(3 + e^x)^4 \cdot e^x \quad \checkmark$$

$$= 5e^x (3 + e^x)^4 \quad \checkmark$$

ii. $y = \frac{e^{2x}}{2x+1}$

2

$$\frac{dy}{dx} = \frac{(2x+1)2e^{2x} - e^{2x}(2)}{(2x+1)^2} \quad \checkmark$$

$$= \frac{4xe^{2x} + 2e^{2x} - 2e^{2x}}{(2x+1)^2}$$

$$= \frac{4xe^{2x}}{(2x+1)^2} \quad \checkmark$$

(b) Find the equation of the tangent to $y = e^{-x}$ at the point (0,1).

2

$$y = e^{-x}$$

$$\frac{dy}{dx} = -e^{-x}$$

$$= -1 \text{ at } (0,1) \quad \checkmark$$

$\therefore$ equation of tangent is:

$$y - 1 = -1(x - 0)$$

$$y = 1 - x \quad \checkmark$$

- (c) Sketch the graph $y = 2e^{-2x} - 2$, showing all important features.

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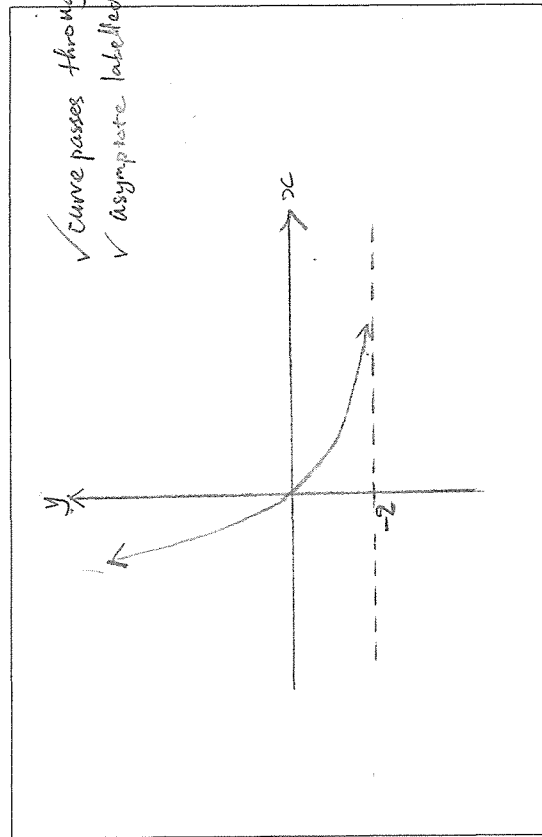
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- (d) Let $f(x) = e^x + k$, where k is a real number. The tangent to the graph of $f(x)$ at the point where $x = a$ passes through the point $(0, 0)$.

- i. Show that the equation of the tangent is $y = e^a x$.

$$f(x) = e^x + k$$

$$f'(x) = e^x$$

$$= e^a \text{ at } x = a$$

The tangent passes through (0,0) ✓

∴ equation of tangent is $y = e^a x$

- ii. Find the value of k in terms of a .

Substitute $x = a$ into $y = e^a x$,

$$y = e^a a$$

$$f(a) = e^a + k$$

$$\therefore e^a + k = e^a a$$

$$k = e^a a - e^a$$

$$= e^a (a - 1) \quad \checkmark$$

Question 2 (18 marks)

Marks

- (a) Evaluate the following:

2

$$\begin{aligned}
 \sum_{k=2}^{10} 4 \times \left(\frac{1}{2}\right)^{k-1} &= 4 \times \frac{1}{2} + 4 \times \left(\frac{1}{2}\right)^2 + 4 \times \left(\frac{1}{2}\right)^3 + \dots \\
 &= \frac{4 \times \frac{1}{2} \left[1 - \left(\frac{1}{2}\right)^9\right]}{1 - \frac{1}{2}} \\
 &= 4 \left[1 - \frac{1}{2^9}\right] \\
 &= 3 \frac{127}{128}
 \end{aligned}$$

- (b) A geometric series has common ratio
- $\frac{1}{p}$
- and limiting sum
- $\frac{1}{1-p}$
- .

2

Find the first term of this series, a , in simplest form.

$$\begin{aligned}
 \frac{1}{1-p} &= \frac{a}{1-\frac{1}{p}} \\
 a(1-p) &= 1 - \frac{1}{p} \\
 a &= \frac{p-1}{p} \times \frac{1}{1-p} \\
 &= -\frac{1}{p}
 \end{aligned}$$

- (c)
- $2m - 8$
- ,
- $2m + 4$
- and
- $5m - 2$
- are successive terms of a geometric sequence. Find the value of
- m
- .

3

$$\begin{aligned}
 \frac{2m+4}{2m-8} &= \frac{5m-2}{2m+4} \\
 (2m+4)^2 &= (5m-2)(2m-8) \\
 4m^2 + 16m + 16 &= 10m^2 - 44m + 16 \\
 6m^2 - 60m &= 0 \\
 6m(m-10) &= 0 \\
 m &= 0 \text{ or } 10
 \end{aligned}$$

- (d) i. Show that
- $\log_{10} 27 + \log_{10} 9 + \log_{10} 3 + \dots$
- is an arithmetic series.

1

$$\begin{aligned}
 T_2 - T_1 &= \log_{10} 9 - \log_{10} 27 = \log_{10} \frac{9}{27} \\
 &= \log_{10} \frac{1}{3} = -\log_{10} 3 \\
 T_3 - T_2 &= \log_{10} 3 - \log_{10} 9 = \log_{10} \frac{3}{9} \\
 &= \log_{10} \frac{1}{3} = -\log_{10} 3
 \end{aligned}$$

- ii. Show that the sum of the first seven terms of this series is 0.

2

$$\begin{aligned}
 S_7 &= \frac{7}{2} [2 \log_{10} 27 + 6(-\log_{10} 3)] \\
 &= \frac{7}{2} [2 \log_{10} 3^3 - 6 \log_{10} 3] \\
 &= \frac{7}{2} [6 \log_{10} 3 - 6 \log_{10} 3] \\
 &= 0
 \end{aligned}$$

- (e) A pattern is made up of a series of concentric circles with centre O . The radius of the first (innermost) circle is 10 cm. The distance between two adjacent concentric circles is 2 cm.

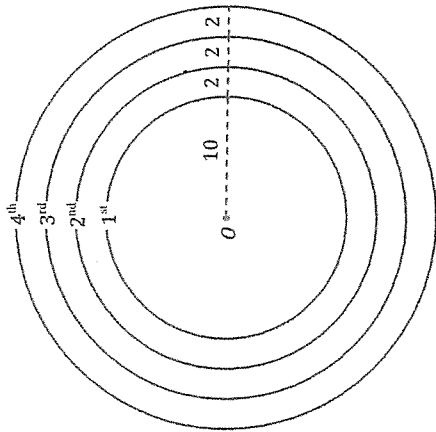


Diagram not drawn to scale

- i. What is circumference of the first circle?

$$20\pi \text{ cm} \quad \checkmark$$

- ii. Show that the circumference, C , of the n^{th} circle is given by $C = 4\pi(4+n)$.

$$\text{Radius: } 10, 12, 14, 16, \dots \quad \text{AP, } a=10, d=2$$

$$\text{Radius of the } n^{\text{th}} \text{ circle}$$

$$= 10 + (n-1)2$$

$$= 8 + 2n \quad \checkmark$$

$$\therefore C = 2\pi(8+2n) \quad \checkmark$$

$$= 4\pi(4+n) \quad \checkmark$$

Alternative solution

$$C: 20\pi, 24\pi, 28\pi, \dots \quad \text{AP, } a=20\pi, d=4\pi$$

$$C = 20\pi + (n-1)4\pi \quad \checkmark$$

$$= 20\pi + 4n\pi - 4\pi \quad \checkmark$$

$$= 16\pi + 4n\pi \quad \checkmark$$

$$= 4\pi(4+n) \quad \checkmark$$

- iii. Show that the total circumference of n circles is given by $L = 2\pi n(9+n)$.

$$C: 20\pi, 24\pi, 28\pi, \dots \quad \text{AP, } a=20\pi, d=4\pi$$

$$\therefore L = \frac{n}{2} [20\pi + 4\pi(4+n)] \quad \checkmark$$

Alternative solution

$$L = \frac{n}{2} [20\pi + (n-1)4\pi] \quad \checkmark$$

$$= \frac{n}{2} [20\pi + 4n\pi - 4\pi] \quad \checkmark$$

$$= \frac{n}{2} [16\pi + 4n\pi] \quad \checkmark$$

$$= \frac{n}{2} [4\pi(4+n)] \quad \checkmark$$

$$= 2\pi n(4+n) \quad \checkmark$$

- iv. If $L = 1584\pi$, find the value of n and hence find the radius of the n^{th} circle.

$$2\pi n(4+n) = 1584\pi$$

$$4n + n^2 = 792 \quad \checkmark$$

$$n^2 + 4n - 792 = 0$$

$$(n-24)(n+33) = 0 \quad \checkmark$$

$$n = 24 \text{ as } n > 0 \quad \checkmark$$

$\therefore$ radius of the 24th circle

$$= 8 + 2n \quad \text{using } C(r)$$

$$= 8 + 2 \times 24 \quad \checkmark$$

$$= 56 \text{ cm} \quad \checkmark$$

$$\therefore r = 56$$

Alternative solution

$$C = 4\pi(4+24)$$

$$= 112\pi$$

$$2\pi r = 112\pi$$

$$\therefore r = 56$$

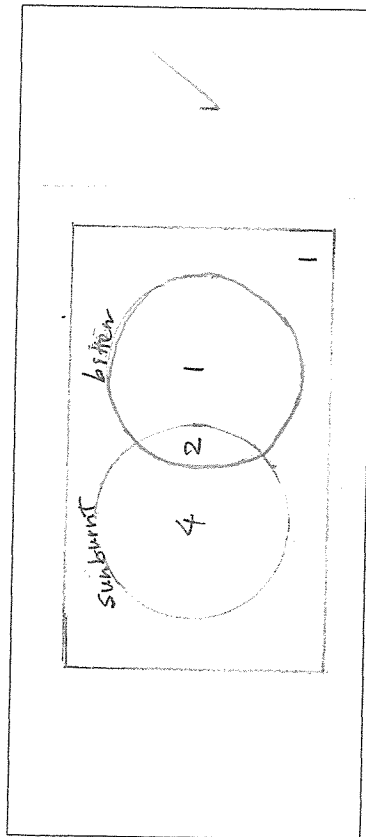
Question 3 (24 marks)

Marks

- (a) 8 members of a rowing team were on a training row.

During the training row 6 members of the team were sunburnt and 2 were sunburnt and bitten by sand-flies. 1 member had nothing happen to him.

- i. Draw a Venn diagram showing this information.



- ii. Hence find the probability that he was sunburnt but not bitten by sand-flies if a student was selected at random from the rowing team.

$$\frac{4}{8} = \frac{1}{2} \quad \checkmark$$

- (b) The chance Janine can solve a probability problem is $\frac{9}{10}$ and the chance Naomi can solve it is $\frac{4}{5}$. If they work on the problem independently, what is the probability that:

- i. They both solve the problem?

$$\frac{9}{10} \times \frac{4}{5} \quad \checkmark$$

$$= \frac{18}{25} \quad \checkmark$$

- ii. At least 1 of them is able to solve the problem?

$$1 - \frac{1}{10} \times \frac{1}{5} \quad \checkmark$$

$$= \frac{49}{50} \quad \checkmark$$

- (c) Four identical balls are numbered 1, 2, 3 and 4 and put into a box.

A ball is randomly drawn from the box, and not returned to the box. A second ball is then randomly drawn from the box.

By drawing a diagram or otherwise, find the probability that:

- i. The first ball drawn is numbered 4 and the second ball drawn is numbered 1.

$$P(4,1) = \frac{1}{4} \times \frac{1}{3}$$

$$= \frac{1}{12} \quad \checkmark$$

- ii. The sum of the numbers on the two balls is 5.

$$\text{Sum} = 5: (1,4), (2,3), (3,2), (4,1)$$

$$P(\text{Sum} = 5) = \frac{4}{12} \times 4$$

$$= \frac{1}{3} \quad \checkmark$$

- iii. The second ball drawn is numbered 1, given that the sum of the numbers on the two balls is 5.

$$P(\text{2nd ball is 1} \mid \text{sum} = 5)$$

$$= \frac{P(\text{2nd ball is 1 and sum} = 5)}{P(\text{sum} = 5)}$$

$$= \frac{\frac{1}{12}}{\frac{1}{3}} \quad \checkmark$$

$$= \frac{1}{4} \quad \checkmark$$

- (d) A and B are events of a sample space. Given that $P(A|B) = p$, $P(B) = p^2$ and $P(A) = p^3$:

- i. Show that $P(A \cap B) = p^3$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$p = \frac{P(A \cap B)}{p^2}$$

$$\therefore P(A \cap B) = p^3 \quad \checkmark$$

- ii. Find $P(B|A)$ in simplest form.

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$= \frac{p^3}{p^3}$$

$$= p^0 \quad \checkmark$$

- (e) From a packet of mixed seed, it was estimated that the probability of any seed planted yielding a white carnation was 0.02.

- i. Calculate the probability that from any two seeds planted there will be at least 1 white carnation.

$$1 - P(\bar{W}\bar{W})$$

$$= 1 - (0.98)^2 \quad \checkmark$$

$$= 0.0396 \quad \checkmark$$

- ii. How many seeds must be planted to be at least 95% certain of obtaining at least 1 white carnation?

$$1 - (0.98)^n = 0.95 \quad \checkmark$$

$$(0.98)^n = 0.05$$

$$n = \frac{\log_{10} 0.05}{\log_{10} 0.98} \quad \checkmark$$

$$= 148.28 \dots$$

$\therefore 149$ seeds must be planted. $\checkmark$

- (f) Reung has invented a game for one person. He throws 2 ordinary dice repeatedly until the sum of the two numbers shown is either a 7 or 9.

If the sum is 9, Reung wins. If the sum is 7, Reung loses. If the sum is any other number, he continues to throw until it is a 7 or 9.

- i. Show that the probability that Reung wins on his first throw is $\frac{1}{9}$.

$$\text{Sum} = 9 : (3,6), (4,5), (5,4), (6,3)$$

$$\therefore P(\text{sum}=9) = \frac{4}{36}$$

$$= \frac{1}{9}$$

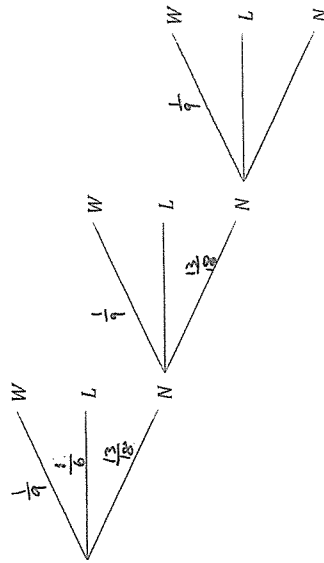
- ii. Calculate the probability that a second throw is needed for Reung to win.

$$\text{Sum} = 7 : (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$$

$$P(\text{sum}=7) = \frac{6}{36} = \frac{1}{6}$$

$$\therefore P(\text{sum is neither 7 nor 9}) = 1 - \frac{1}{9} - \frac{1}{6} = \frac{13}{18}$$

- iii. By completing the tree diagram or otherwise, find the probability that Reung wins on his first, second or third throw.



$$\begin{aligned} &P(\text{wins on 1st, 2nd or 3rd throw}) \\ &= P(\text{wins on 1st throw}) + P(\text{wins on 2nd throw}) + P(\text{wins on 3rd throw}) \\ &= \frac{1}{9} + \frac{13}{18} \times \frac{1}{9} + \frac{13}{18} \times \frac{13}{18} \times \frac{1}{9} \\ &= \frac{1}{9} \left[1 + \frac{13}{18} + \left(\frac{13}{18} \right)^2 \right] \\ &= 0.2493 \end{aligned}$$

- iv. Calculate the probability that Reung wins the game. That is, that Reung wins on his first or second or third or 4th throw.

$$\frac{1}{9} \left[1 + \frac{13}{18} + \left(\frac{13}{18} \right)^2 + \left(\frac{13}{18} \right)^3 + \dots \right]$$

$$= \frac{1}{9} \times \frac{1}{1 - \frac{13}{18}}$$

$$= 0.24$$